

JomCode Coding Bootcamp

Database modeling, SQL and Supabase

Understand, predict, type, run, then practice.

What you will be able to do

- 01 Model users, workshops and registrations using keys and constraints.
- 02 Distinguish PostgreSQL, SQLAlchemy and Supabase.
- 03 Create a Supabase project and keep its connection string private.

Plain meaning → technical view → use

PLAIN MEANING

Store related records so their connections stay valid.

TECHNICAL VIEW

PostgreSQL tables use primary keys, foreign keys and constraints; Supabase hosts the database.

WHEN YOU USE IT

Connect users, workshops and registrations without duplicate free-form records.

TIP

Draw tables and their keys before opening the database dashboard.

Worked example 1: Safe SQL practice data



chapter_15_database_modeling_sql_and_supabase.py

```
1 WITH workshops(id, title, capacity) AS (  
2   VALUES ('w1', 'Python', 20), ('w2', 'SQL', 5)  
3 )  
4 SELECT title, capacity FROM workshops  
5 WHERE capacity ≥ 10 ORDER BY title;
```

PREDICT Which clause filters before presentation?

Example 1: output and reason

TERMINAL >_ EXPECTED OUTPUT

One row: Python,20. The CTE changes no persisted data.

WHY THIS HAPPENS

WHERE selects matching rows. ORDER BY makes presentation deterministic.

Exercise 1: Entity diagram

BUILD

Describe the three tables and their keys. Include deletion and duplicate policies.

CHECK

Workshop removal cascades its registrations. Email is unique. Each workshop/participant pair is unique.

IF STUCK

A relationship and a duplicate rule are different constraints.

Worked example 2: Aggregating a relationship



chapter_15_database_modeling_sql_and_supabase.py

```
1 WITH registrations(workshop_id, participant_id) AS (  
2     VALUES ('w1', 'p1'), ('w1', 'p2'), ('w2', 'p1')  
3 )  
4 SELECT workshop_id, COUNT(*) AS registered_count  
5 FROM registrations GROUP BY workshop_id ORDER BY workshop_id;
```

PREDICT Can one participant join several workshops?

Example 2: output and reason

TERMINAL >_ EXPECTED OUTPUT

```
w1 has 2 registrations. w2 has 1.
```

WHY THIS HAPPENS

Yes. The unique constraint applies to each workshop/participant pair.

Exercise 2: Literal search and pagination

BUILD

Write a safe CTE query listing only titles containing Python, ordered by title and id, first two records.

CHECK

Two matching records maximum. Stable ordering. No concatenated user SQL.

IF STUCK

The fixed practice literal is safe. Runtime input uses driver parameters.

Worked example 3: Explicit transaction boundary



chapter_15_database_modeling_sql_and_supabase.py

```
1  -- Reference pattern, after the guided schema exists.
2  BEGIN;
3  SELECT id, capacity FROM bootcamp.workshops
4  WHERE id = :workshop_id FOR UPDATE;
5  -- Count and insert use SQLAlchemy bound parameters in the repository.
6  COMMIT;
```

PREDICT Why is a separate count then insert unsafe without a lock?

Example 3: output and reason

TERMINAL >_ EXPECTED OUTPUT

```
The workshop row serializes competing capacity changes. :workshop_id is a
driver-bound parameter, not SQL Editor syntax.
```

WHY THIS HAPPENS

Two requests could both see the last seat. The later repository chapter implements this whole transaction.

Exercise 3: Connection evidence

BUILD

During guided milestone 2, verify the Supabase backend connection and restricted role. Record only nonsecret evidence.

CHECK

Connection succeeds over TLS. Runtime role can access bootcamp but cannot create tables. No URI in a screenshot.

IF STUCK

Start with the Connect panel session-pooler URI rather than guessing a hostname.

Checkpoint

PROVE IT

Learners can explain the relational model and find a private Supabase connection string. Live backend connectivity follows in guided milestone 2.